

1) Планирование адресного пространства

Предлагаем следующую схему:

Подсеть	Назначение	Адрес сети	Маска	Gateway (R1)
LAN1	ПК1, ПК2	192.168.10.0	/24	192.168.10.1
LAN2	ПК3, ПК4	192.168.20.0	/24	192.168.20.1
LAN3	R1–R2 линк	192.168.30.0	/30	—

На LAN3 маленькая /30, т.к. это только линк между R1 и R2.

2) Адреса на маршрутизаторах (статически)

R1

conf t

```
interface GigabitEthernet0/0 ! LAN1
```

```
ip address 192.168.10.1 255.255.255.0
```

```
no shut
```

```
interface GigabitEthernet0/2 ! LAN2
```

```
ip address 192.168.20.1 255.255.255.0
```

```
no shut
```

```
interface GigabitEthernet0/1 ! link to R2 (LAN3)
```

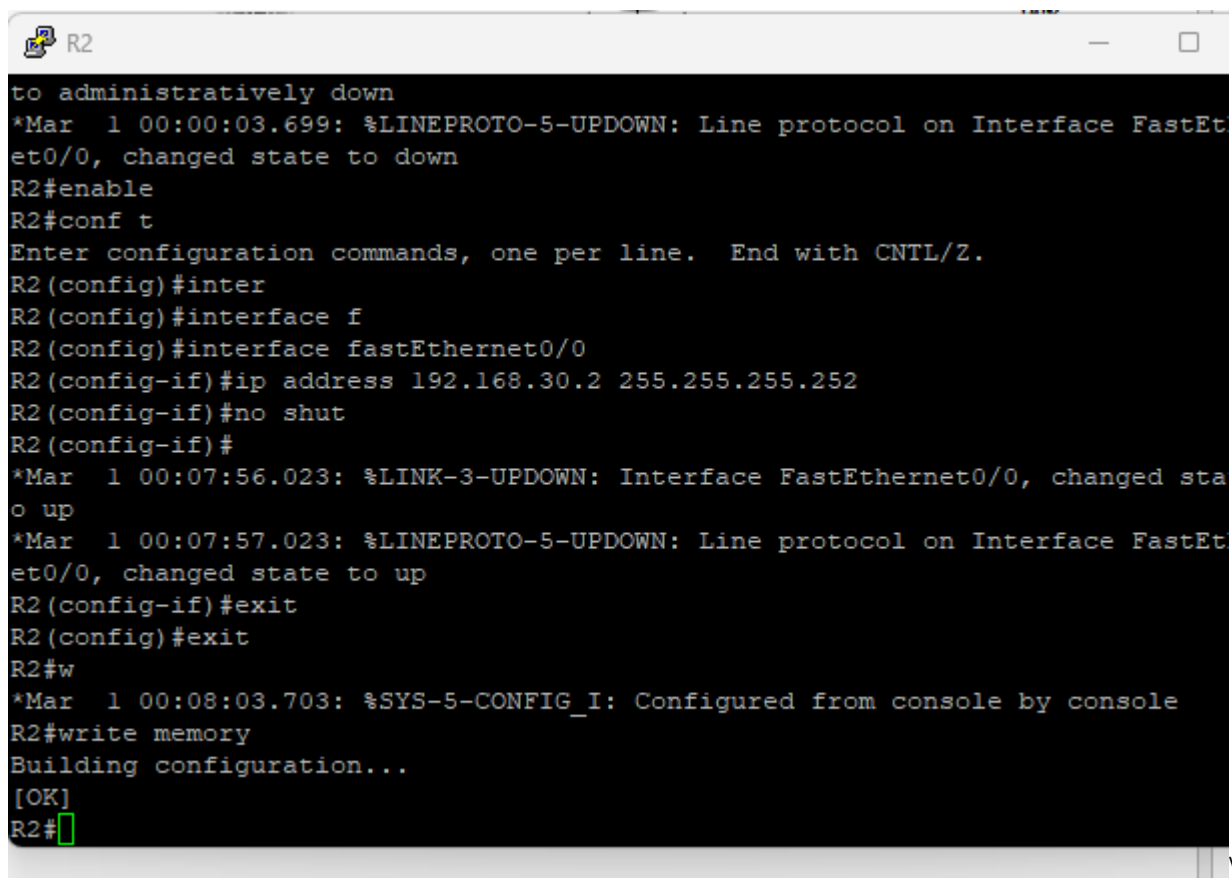
```
ip address 192.168.30.1 255.255.255.252
```

```
no shut
```

R2 (DHCP сервер)

conf t

```
interface GigabitEthernet0/0 ! link to R1
ip address 192.168.30.2 255.255.255.252
no shut
```



```
R2
to administratively down
*Mar  1 00:00:03.699: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEt
et0/0, changed state to down
R2#enable
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#inter
R2(config)#interface f
R2(config)#interface fastEthernet0/0
R2(config-if)#ip address 192.168.30.2 255.255.255.252
R2(config-if)#no shut
R2(config-if)#
*Mar  1 00:07:56.023: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed sta
o up
*Mar  1 00:07:57.023: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEt
et0/0, changed state to up
R2(config-if)#exit
R2(config)#exit
R2#w
*Mar  1 00:08:03.703: %SYS-5-CONFIG_I: Configured from console by console
R2#write memory
Building configuration...
[OK]
R2#
```

3) DHCP НА R2 ДЛЯ LAN1 и LAN2

nb! СЕРВЕР ТОЛЬКО НА R2

Отключаем выдачу адреса на линке LAN3

```
conf t
```

```
interface GigabitEthernet0/0
```

```
no ip dhcp client request
```

DHCP pool для LAN1

```
ip dhcp pool LAN1
```

```
network 192.168.10.0 255.255.255.0
```

```
default-router 192.168.10.1
```

```
dns-server 8.8.8.8
```

DHCP pool для LAN2

```
ip dhcp pool LAN2  
network 192.168.20.0 255.255.255.0  
default-router 192.168.20.1  
dns-server 8.8.8.8
```

Исключаем адреса шлюзов

```
ip dhcp excluded-address 192.168.10.1  
ip dhcp excluded-address 192.168.20.1
```

```
R2#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
R2(config)#ip dhcp pool LAN1  
R2(dhcp-config)#network 192.168.10.0 255.255.255.0  
R2(dhcp-config)#default-ro  
R2(dhcp-config)#default-router 192.168.10.1  
R2(dhcp-config)#dns-  
R2(dhcp-config)#dns-server 8.8.8.8  
R2(dhcp-config)#exit  
R2(config)#exit  
R2#  
*Mar  1 00:41:40.259: %SYS-5-CONFIG_I: Configured from console by console  
R2#write memory  
Building configuration...  
[OK]  
R2#
```

```
R2(config)#ip dhcp pool LAN2  
R2(dhcp-config)#network 192.168.10.0 255.255.255.0  
R2(dhcp-config)#default-route  
R2(dhcp-config)#default-router 192.168.10.1  
R2(dhcp-config)#dns-se  
R2(dhcp-config)#dns-server 8.8.8.8  
R2(dhcp-config)#exit  
R2(config)#ip dhcp pool LAN2  
R2(dhcp-config)#network 192.168.20.0 255.255.255.0  
R2(dhcp-config)#default-router 192.168.20.1  
R2(dhcp-config)#dns-server 8.8.8.8  
R2(dhcp-config)#exit  
R2(config)#ip dhcp excluded-address 192.168.10.1  
R2(config)#ip dhcp excluded-address 192.168.20.1  
R2(config)#
```

nb! 4) DHCP RELAY НА R1

Так как DHCP-сервер в другой подсети, обязательно:

На интерфейсе LAN1

```
interface GigabitEthernet0/0
```

```
ip helper-address 192.168.30.2
```

На интерфейсе LAN2

```
interface GigabitEthernet0/2  
ip helper-address 192.168.30.2
```

```
R1#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
R1(config)#inter  
R1(config)#interface f  
R1(config)#interface fastEthernet0/0  
R1(config-if)#ip helper-address 192.168.30.2  
R1(config-if)#exit  
R1(config)#exit  
R1#  
*Mar  1 00:25:09.407: %SYS-5-CONFIG_I: Configured from console by console  
R1#write memory  
Building configuration...  
[OK]  
R1#co  
R1#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
R1(config)#inter  
R1(config)#interface f  
R1(config)#interface fastEthernet2/0  
R1(config-if)#ip helper-address 192.168.30.2  
R1(config-if)#exit  
R1(config)#exit  
R1#  
*Mar  1 00:27:36.511: %SYS-5-CONFIG_I: Configured from console by console  
R1#write memory  
Building configuration...  
[OK]  
R1#
```

5) Статическая маршрутизация

На R1 — маршрут к DHCP серверу уже есть (прямое подключение)

ничего добавлять не нужно

На R2 — маршруты к LAN1 и LAN2

```
ip route 192.168.10.0 255.255.255.0 192.168.30.1
```

```
ip route 192.168.20.0 255.255.255.0 192.168.30.1
```

```
R2(config)#ip route 192.168.10.0 255.255.255.0 192.168.30.1  
R2(config)#ip route 192.168.20.0 255.255.255.0 192.168.30.1  
R2(config)#
```

6) Конфигурация VPC (в GNS3)

На всех ПК просто включить DHCP:

```
PC1 - PuTTY  
PC1> ip dhcp  
DDORA IP 192.168.10.2/24 GW 192.168.10.1  
PC1> █
```

```
PC2> ip dhcp  
DDORA IP 192.168.10.3/24 GW 192.168.10.1  
PC2> █
```

```
PC3> ip dhcp  
DDORA IP 192.168.20.2/24 GW 192.168.20.1  
PC3> █
```

```
PC4> ip dhcp  
DDORA IP 192.168.20.3/24 GW 192.168.20.1  
PC4> █
```

ip dhcp

Проверить:

show ip

The image displays four terminal windows, each representing a different PC in a network configuration. Each window shows the execution of DHCP and IP configuration commands, followed by a 'show ip' command that displays the resulting network parameters.

PC2 - PuTTY

```
For more information, please visit wiki.freecode.com.cn.  
Press '?' to get help.  
Executing the startup file  
  
PC2> ip dhcp  
DDORA IP 192.168.10.3/24 GW 192.168.10.1  
  
PC2> show ip  
NAME : PC2[1]  
IP/MASK : 192.168.10.3/24  
GATEWAY : 192.168.10.1  
DNS : 8.8.8.8  
DHCP SERVER : 192.168.30.2  
DHCP LEASE : 86288, 86400/43200/75600  
MAC : 00:50:79:66:68:01  
LPORT : 23827  
RHOST:PORT : 127.0.0.1:23828  
MTU : 1500  
PC2>
```

PC3 - PuTTY

```
For more information, please visit wiki.freecode.com.cn.  
Press '?' to get help.  
Executing the startup file  
  
PC3> ip dhcp  
DDORA IP 192.168.20.2/24 GW 192.168.20.1  
  
PC3> show ip  
NAME : PC3[1]  
IP/MASK : 192.168.20.2/24  
GATEWAY : 192.168.20.1  
DNS : 8.8.8.8  
DHCP SERVER : 192.168.30.2  
DHCP LEASE : 86342, 86400/43200/75600  
MAC : 00:50:79:66:68:02  
LPORT : 23835  
RHOST:PORT : 127.0.0.1:23836  
MTU : 1500  
PC3>
```

PC1 - PuTTY

```
PC1> ip dhcp  
DDORA IP 192.168.10.2/24 GW 192.168.10.1  
  
PC1> show ip  
NAME : PC1[1]  
IP/MASK : 192.168.10.2/24  
GATEWAY : 192.168.10.1  
DNS : 8.8.8.8  
DHCP SERVER : 192.168.30.2  
DHCP LEASE : 86214, 86400/43200/75600  
MAC : 00:50:79:66:68:00  
LPORT : 23825  
RHOST:PORT : 127.0.0.1:23826  
MTU : 1500  
PC1>
```

PC4 - PuTTY

```
For more information, please visit wiki.freecode.com.cn.  
Press '?' to get help.  
Executing the startup file  
  
PC4> ip dhcp  
DDORA IP 192.168.20.3/24 GW 192.168.20.1  
  
PC4> show ip  
NAME : PC4[1]  
IP/MASK : 192.168.20.3/24  
GATEWAY : 192.168.20.1  
DNS : 8.8.8.8  
DHCP SERVER : 192.168.30.2  
DHCP LEASE : 86343, 86400/43200/75600  
MAC : 00:50:79:66:68:03  
LPORT : 23837  
RHOST:PORT : 127.0.0.1:23838  
MTU : 1500  
PC4>
```

ПК1 → 192.168.10.x

ПК2 → 192.168.10.x

ПК3 → 192.168.20.x

ПК4 → 192.168.20.x

7) Проверка связности (ping)

```
DDORA IP 192.168.10.3/24 GW 192.168.10.1
PC2> show ip
NAME       : PC2[1]
IP/MASK    : 192.168.10.3/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.30.2
DHCP LEASE : 86288, 86400/43200/75600
MAC        : 00:50:79:66:68:01
LPORT     : 23827
RHOST:PORT : 127.0.0.1:23828
MTU        : 1500
PC2> ping 192.168.20.3
84 bytes from 192.168.20.3 icmp_seq=1 ttl=63 time=29.284 ms
84 bytes from 192.168.20.3 icmp_seq=2 ttl=63 time=15.378 ms
84 bytes from 192.168.20.3 icmp_seq=3 ttl=63 time=15.295 ms
84 bytes from 192.168.20.3 icmp_seq=4 ttl=63 time=25.038 ms
84 bytes from 192.168.20.3 icmp_seq=5 ttl=63 time=16.268 ms
PC2>

DDORA IP 192.168.20.2/24 GW 192.168.20.1
PC3> show ip
NAME       : PC3[1]
IP/MASK    : 192.168.20.2/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.30.2
DHCP LEASE : 86342, 86400/43200/75600
MAC        : 00:50:79:66:68:02
LPORT     : 23835
RHOST:PORT : 127.0.0.1:23836
MTU        : 1500
PC3> ping 192.168.30.2
84 bytes from 192.168.30.2 icmp_seq=1 ttl=254 time=14.
84 bytes from 192.168.30.2 icmp_seq=2 ttl=254 time=15.
84 bytes from 192.168.30.2 icmp_seq=3 ttl=254 time=15.
84 bytes from 192.168.30.2 icmp_seq=4 ttl=254 time=15.
84 bytes from 192.168.30.2 icmp_seq=5 ttl=254 time=16.
PC3>

DDORA IP 192.168.10.2/24 GW 192.168.10.1
PC1> show ip
NAME       : PC1[1]
IP/MASK    : 192.168.10.2/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.30.2
DHCP LEASE : 86214, 86400/43200/75600
MAC        : 00:50:79:66:68:00
LPORT     : 23825
RHOST:PORT : 127.0.0.1:23826
MTU        : 1500
PC1> ping 192.168.20.2
84 bytes from 192.168.20.2 icmp_seq=1 ttl=63 time=29.604 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=63 time=14.654 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=63 time=16.982 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=63 time=24.885 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=63 time=15.315 ms
PC1> ping 192.168.

Executing the startup file
PC4> ip dhcp
DDORA IP 192.168.20.3/24 GW 192.168.20.1
PC4> show ip
NAME       : PC4[1]
IP/MASK    : 192.168.20.3/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.30.2
DHCP LEASE : 86343, 86400/43200/75600
MAC        : 00:50:79:66:68:03
LPORT     : 23837
RHOST:PORT : 127.0.0.1:23838
MTU        : 1500
PC4> GET / HTTP/1.1
Bad command: "GET / HTTP/1.1". Use ? for help.
PC4>
```

ПК1 → ПК3

ПК2 → ПК4

ПК3 → R2

ПК1 → R1

8) Wireshark — анализ DHCP диалога

Захват с Standard input [PC1 Ethernet0 to Layer2Switch-1 Ethernet0]

No.	Time	Source	Destination	Protocol	Length	Info
79	111.583925	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x970a3f63
82	112.584029	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x970a3f63
84	115.584110	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x970a3f63
574	825.375918	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x32bc4414
577	826.376064	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x32bc4414
578	827.101307	192.168.10.1	192.168.10.2	DHCP	342	DHCP Offer - Transaction ID 0x32bc4414
579	827.101325	192.168.10.1	192.168.10.2	DHCP	342	DHCP Offer - Transaction ID 0x32bc4414
581	829.376116	0.0.0.0	255.255.255.255	DHCP	406	DHCP Request - Transaction ID 0x32bc4414
582	829.394739	192.168.10.1	192.168.10.2	DHCP	342	DHCP ACK - Transaction ID 0x32bc4414
637	905.156949	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0xc740456a
639	906.156311	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0xc740456a

Захват с Standard input [Layer2Switch-1 Ethernet2 to R1 FastEthernet0/0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

dhcpc

No.	Time	Source	Destination	Protocol	Length	Info
89	107.585113	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x970a3f63
92	108.584701	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x970a3f63
94	111.584899	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x970a3f63
666	821.376610	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x32bc4414
669	822.376966	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x32bc4414
670	823.101302	192.168.10.1	192.168.10.2	DHCP	342	DHCP Offer - Transaction ID 0x32bc4414
671	823.101320	192.168.10.1	192.168.10.2	DHCP	342	DHCP Offer - Transaction ID 0x32bc4414
673	825.376844	0.0.0.0	255.255.255.255	DHCP	406	DHCP Request - Transaction ID 0x32bc4414
674	825.394769	192.168.10.1	192.168.10.2	DHCP	342	DHCP ACK - Transaction ID 0x32bc4414
739	901.157301	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0xc740456a
741	902.156657	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0xc740456a
743	902.865399	192.168.10.1	192.168.10.3	DHCP	342	DHCP Offer - Transaction ID 0xc740456a
744	902.865425	192.168.10.1	192.168.10.3	DHCP	342	DHCP Offer - Transaction ID 0xc740456a
746	905.156767	0.0.0.0	255.255.255.255	DHCP	406	DHCP Request - Transaction ID 0xc740456a
747	905.169341	192.168.10.1	192.168.10.3	DHCP	342	DHCP ACK - Transaction ID 0xc740456a

Захват с Standard input [R1 FastEthernet1/0 to R2 FastEthernet0/0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

dhcpc

No.	Time	Source	Destination	Protocol	Length	Info
27	99.872497	192.168.10.1	192.168.30.2	DHCP	406	DHCP Discover - Transaction ID 0x970a3f63
28	102.880501	192.168.10.1	192.168.30.2	DHCP	406	DHCP Discover - Transaction ID 0x970a3f63
195	812.665779	192.168.10.1	192.168.30.2	DHCP	406	DHCP Discover - Transaction ID 0x32bc4414
197	813.672566	192.168.10.1	192.168.30.2	DHCP	406	DHCP Discover - Transaction ID 0x32bc4414
199	814.385064	192.168.30.2	192.168.10.1	DHCP	342	DHCP Offer - Transaction ID 0x32bc4414
200	814.385085	192.168.30.2	192.168.10.1	DHCP	342	DHCP Offer - Transaction ID 0x32bc4414
201	816.669988	192.168.10.1	192.168.30.2	DHCP	406	DHCP Request - Transaction ID 0x32bc4414
202	816.671388	192.168.30.2	192.168.10.1	DHCP	342	DHCP ACK - Transaction ID 0x32bc4414
220	892.450415	192.168.10.1	192.168.30.2	DHCP	406	DHCP Discover - Transaction ID 0xc740456a
222	893.446616	192.168.10.1	192.168.30.2	DHCP	406	DHCP Discover - Transaction ID 0xc740456a
225	894.140994	192.168.30.2	192.168.10.1	DHCP	342	DHCP Offer - Transaction ID 0xc740456a
226	894.141018	192.168.30.2	192.168.10.1	DHCP	342	DHCP Offer - Transaction ID 0xc740456a
227	896.444555	192.168.10.1	192.168.30.2	DHCP	406	DHCP Request - Transaction ID 0xc740456a
228	896.445219	192.168.30.2	192.168.10.1	DHCP	342	DHCP ACK - Transaction ID 0xc740456a
243	958.205043	192.168.20.1	192.168.30.2	DHCP	406	DHCP Discover - Transaction ID 0xbd18c00f
245	959.211446	192.168.20.1	192.168.30.2	DHCP	406	DHCP Discover - Transaction ID 0xbd18c00f
247	959.923801	192.168.30.2	192.168.20.1	DHCP	342	DHCP Offer - Transaction ID 0xbd18c00f
248	959.923823	192.168.30.2	192.168.20.1	DHCP	342	DHCP Offer - Transaction ID 0xbd18c00f
249	962.210380	192.168.20.1	192.168.30.2	DHCP	406	DHCP Request - Transaction ID 0xbd18c00f
250	962.218082	192.168.30.2	192.168.20.1	DHCP	342	DHCP ACK - Transaction ID 0xbd18c00f
251	962.804004	192.168.20.1	192.168.30.2	DHCP	406	DHCP Discover - Transaction ID 0x98eae974
254	963.810188	192.168.20.1	192.168.30.2	DHCP	406	DHCP Discover - Transaction ID 0x98eae974
256	964.521752	192.168.30.2	192.168.20.1	DHCP	342	DHCP Offer - Transaction ID 0x98eae974
257	964.521766	192.168.30.2	192.168.20.1	DHCP	342	DHCP Offer - Transaction ID 0x98eae974
259	966.807468	192.168.20.1	192.168.30.2	DHCP	406	DHCP Request - Transaction ID 0x98eae974
260	966.815649	192.168.30.2	192.168.20.1	DHCP	342	DHCP ACK - Transaction ID 0x98eae974

DHCP DISCOVER (Client → Broadcast 255.255.255.255)

DHCP OFFER (Server → Client)

DHCP REQUEST (Client → Broadcast)

DHCP ACK (Server → Client)

– Discover: клиент ищет сервер

– Offer: сервер предлагает адрес

– Request: клиент соглашается

– АСК: сервер подтверждает и применяет

9) Сохранение конфигураций устройств

На маршрутизаторах

copy running-config